AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

==

THE SITE

Location	Aligned Data Centers DFW-04, TX -- station KTKI, America/Chicago
Building	OSM 1486780799 Aligned Data Centers DFW-04
Facade gap	not applicable -- one building, no second facade
Weather record	43,718 real hourly records from KTKI
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-02-26 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 24 hours with 2 mode changes. 9 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	10 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	10 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 9 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	7.780	18.0	8.890	4.244
01	FREE	yes	-	8.330	18.0	8.890	3.830
02	FREE	yes	-	8.340	18.0	8.330	3.015
03	FREE	yes	-	8.890	18.0	7.780	3.079
04	FREE	yes	-	8.890	18.0	7.220	2.738

05	FREE	yes	-	8.890	18.0	7.220	2.475
06	FREE	yes	-	9.440	18.0	7.220	2.627
07	FREE	yes	-	10.000	18.0	7.220	2.080
08	FREE	yes	-	12.770	18.0	7.780	3.145
09	FREE	yes	-	15.560	18.0	8.890	4.179
10	mechanical	no	dry-bulb	18.330	18.0	10.000	5.583
11	mechanical	no	dry-bulb	20.000	18.0	11.110	5.704
12	mechanical	no	dry-bulb	21.670	18.0	12.220	6.206
13	mechanical	no	dry-bulb	21.660	18.0	13.330	6.125
14	mechanical	no	dry-bulb	18.900	18.0	13.890	4.155
15	mechanical	yes	switch budget	17.230	18.0	13.890	3.115
16	mechanical	yes	switch budget	16.110	18.0	13.890	2.665
17	mechanical	yes	switch budget	15.550	18.0	13.890	3.089
18	mechanical	yes	switch budget	14.440	18.0	13.330	3.058
19	mechanical	yes	switch budget	13.330	18.0	12.220	2.979
20	mechanical	yes	switch budget	12.780	18.0	12.220	3.595
21	mechanical	yes	switch budget	12.220	18.0	10.000	4.094
22	mechanical	yes	switch budget	12.220	18.0	10.000	4.566
23	mechanical	yes	switch budget	12.230	18.0	9.440	4.280

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.780 C, which is 10.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.244 C, and the margin is measured, not chosen: 4.244 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.330 C, which is 9.670 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.830 C, and the margin is measured, not chosen: 3.830 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.340 C, which is 9.660 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.015 C, and the margin is measured, not chosen: 3.015 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.890 C, which is 9.110 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.079 C, and the margin is measured, not chosen: 3.079 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.890 C, which is 9.110 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.738 C, and the margin is measured, not chosen: 2.738 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.890 C, which is 9.110 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.476 C, and the margin is measured, not chosen: 2.476 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.440 C, which is 8.560 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.627 C, and the margin is measured, not chosen: 2.627 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.000 C, which is 8.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.080 C, and the margin is measured, not chosen: 2.080 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.770 C, which is 5.230 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.145 C, and the margin is measured, not chosen: 3.145 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.560 C, which is 2.440 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.179 C, and the margin is measured, not chosen: 4.179 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.330 C against a 18.0 C limit, so it fails by 0.330 C. It would take a limit 0.330 C higher, or a bound 0.330 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.000 C against a 18.0 C limit, so it fails by 2.000 C. It would take a limit 2.000 C higher, or a bound 2.000 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.670 C against a 18.0 C limit, so it fails by 3.670 C. It would take a limit 3.670 C higher, or a bound 3.670 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.660 C against a 18.0 C limit, so it fails by 3.660 C. It would take a limit 3.660 C higher, or a bound 3.660 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.900 C against a 18.0 C limit, so it fails by 0.900 C. It would take a limit 0.900 C higher, or a bound 0.900 C tighter, to change this hour.

15:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.230 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.110 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.550 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.440 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.330 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.780 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.220 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.220 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.230 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

Generated by AGENTIC-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.